Food Waste Monitoring System

1. Problem Statement

Develop a Python program to monitor food waste over several days by recording the number of meals prepared, meals served, and meals left over each day. Calculate the total meals prepared, served, and wasted, determine the daily food waste percentage, and identify the days on which food waste exceeds the acceptable limit.

2. Objectives

- To record the number of meals prepared and served each day.
- To calculate the leftover meals for each day.
- To calculate the percentage of food wasted.
- To identify days when the waste percentage exceeds the acceptable limit.
- To promote effective food management and reduce food wastage.

3. Assumptions

1. The program monitors food waste for seven days.
2. The number of meals prepared is greater than zero each day.
3. Meals served cannot exceed meals prepared.
4. Leftover meals are calculated by subtracting meals served from meals prepared.
5. The acceptable food waste limit is assumed to be 10%.
6. All meal quantities are entered as non-negative integers.

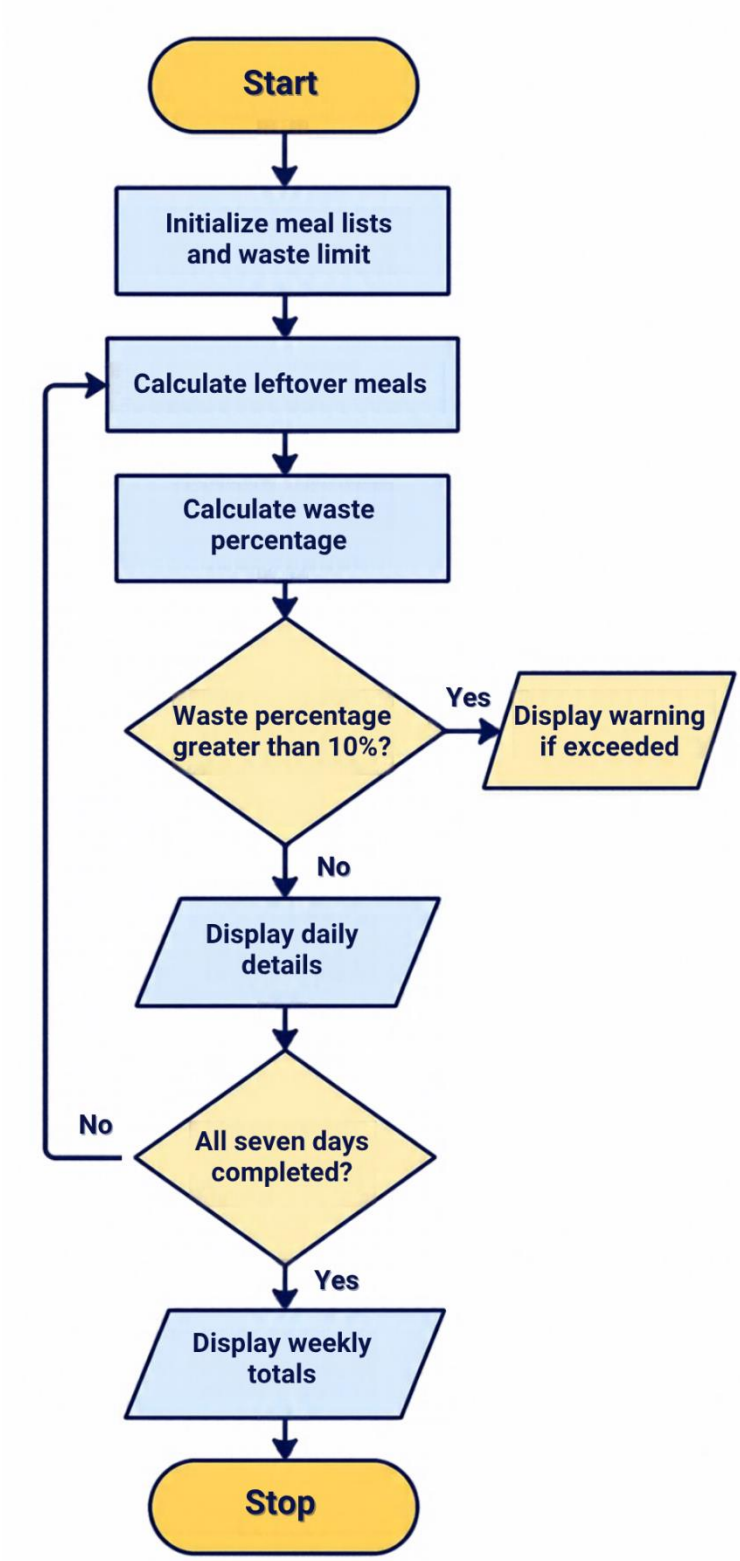
4. Algorithm

1. Start the program.
2. Initialize lists containing the meals prepared and served for seven days.
3. Set the acceptable waste percentage to 10%.
4. Initialize variables to store total meals prepared, served, and wasted.
5. Repeat for each day:

- Calculate leftover meals.
- Calculate the waste percentage.
- Display the daily details.
- Check whether the waste percentage exceeds 10%.
- Display a warning if the limit is exceeded.

6. Calculate and display the weekly totals.
7. Stop the program.

5. Flowchart



6. Pseudocode

START

```
SET prepared = [100, 120, 110, 130, 100, 150, 120]
SET served = [95, 100, 105, 110, 92, 130, 100]
SET limit = 10

SET total_prepared = 0
SET total_served = 0
SET total_waste = 0

FOR each day from 0 to 6:
    leftover = prepared[day] - served[day]
    waste_percentage = (leftover / prepared[day]) * 100

    DISPLAY day, prepared, served, leftover
    DISPLAY waste_percentage

    IF waste_percentage > limit:
        DISPLAY "Waste limit exceeded"
    ELSE:
        DISPLAY "Waste within limit"

    ADD prepared[day] to total_prepared
    ADD served[day] to total_served
    ADD leftover to total_waste
```

DISPLAY total_prepared

DISPLAY total_served

DISPLAY total_waste

STOP

7. Python Actual Code

```
# Food Waste Monitoring System

prepared = [100, 120, 110, 130, 100, 150, 120]
served = [95, 100, 105, 110, 92, 130, 100]

limit = 10

total_prepared = 0
total_served = 0
total_waste = 0

print("FOOD WASTE MONITORING SYSTEM")
print("-" * 65)

for i in range(len(prepared)):
    leftover = prepared[i] - served[i]
    waste_percentage = (leftover / prepared[i]) * 100
```

```
    print(f"\nDay {i + 1}")
    print("Meals Prepared:", prepared[i])
    print("Meals Served:", served[i])
    print("Meals Left Over:", leftover)
    print(f"Waste Percentage: {waste_percentage:.2f}%")

    if waste_percentage > limit:
        print("Status: Waste limit exceeded!")
    else:
        print("Status: Waste within limit.")

    total_prepared += prepared[i]
    total_served += served[i]
    total_waste += leftover

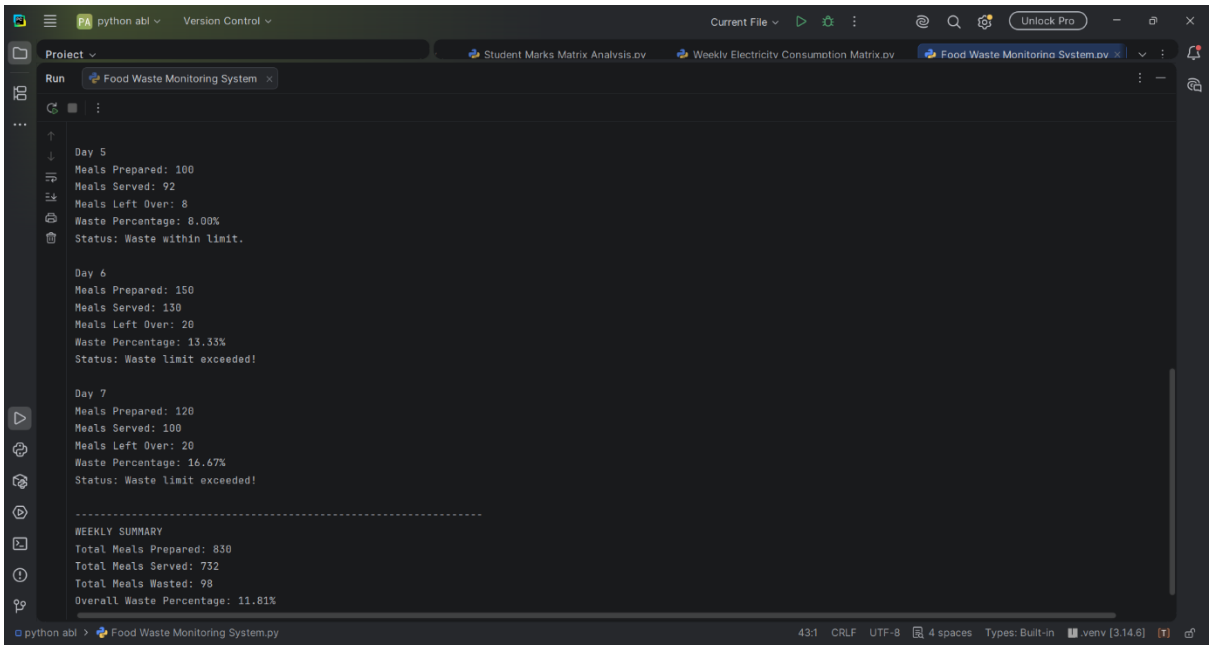
print("\n" + "-" * 65)
print("WEEKLY SUMMARY")
print("Total Meals Prepared:", total_prepared)
print("Total Meals Served:", total_served)
print("Total Meals Wasted:", total_waste)

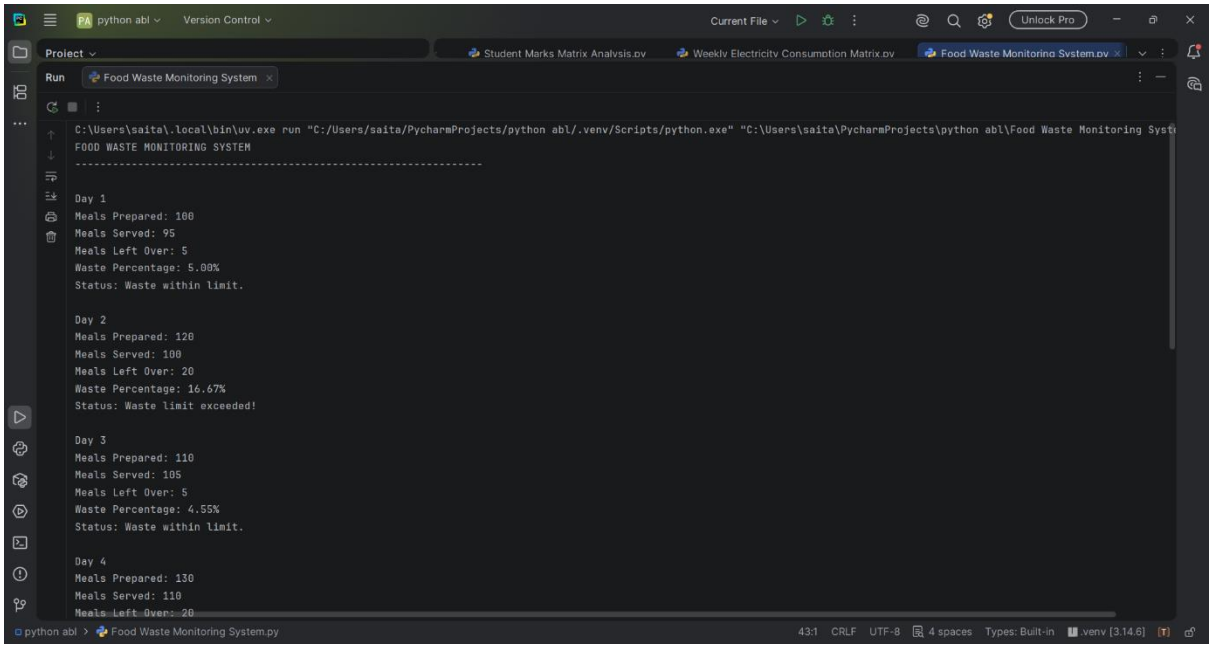
weekly_percentage = (total_waste / total_prepared) * 100
print(f"Overall Waste Percentage: {weekly_percentage:.2f}%")
```

8. Explanation of the Code

1. Lists: prepared and served store the daily meal quantities for seven days.
2. Waste limit: The variable limit = 10 sets the acceptable waste percentage to 10%.
3. Loop: for i in range(len(prepared)) processes each day one by one.
4. Leftover calculation: leftover = prepared[i] - served[i] calculates the meals not served.
5. Waste percentage: The program divides leftover meals by prepared meals and multiplies by 100.
6. Conditional statement: The if statement checks whether the waste percentage exceeds 10%.
7. Accumulation: The += operator adds daily quantities to the weekly totals.
8. Weekly summary: The program displays the total meals prepared, served, wasted, and the overall waste percentage.

9. Output





10. Conclusion

The Food Waste Monitoring System successfully calculates daily leftover meals and food waste percentages using Python lists, loops, arithmetic operations, and conditional statements. It identifies days when food waste exceeds the acceptable 10% limit and calculates the overall weekly waste percentage. This program helps monitor food consumption and encourages better meal planning to reduce food wastage.